LESSON PLAN

AIRCRAFT WIRING PRACTICES (ELECTRICAL WIRING AND INTERFACE SYSTEM) (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Auto rotation, ground resonance, vortex ring, blade sailing
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Auto rotation, ground resonance, vortex ring, blade sailing.
5. Specific Objectives	12.1.7.1 Trainee will be able to explain auto rotation, ground resonance, vortex ring, blade sailing.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight simulation scenarios demonstrating auto rotation entry, ground resonance excitation, vortex ring state recovery, and blade sailing conditions. Safety: Ensure simulator controls are reset between scenarios. Instructor supervision: Monitor trainee interpretation of flight data and recovery procedures.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of auto rotation in helicopter flight?
(b) Define ground resonance and identify the component most susceptible to it.
(c) How does the rotor disc behave during vortex ring state?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principles of Auto Rotation • Energy conversion in descent • Control inputs during entry • Recovery procedures • Limitations of auto rotation	L & Ds	PPT, IB	15 min
(b)	• Ground Resonance Mechanics • Blade Sailing Phenomenon • Vortex Ring State Dynamics • Prevention and Recovery • System limitations	L & Ds	PPT, PDF	20 min
(c)	• Comparative Analysis of Hazards • Emergency Procedures Review • Pilot Decision Making • Maintenance Implications • Operational Constraints	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between ground resonance and blade sailing.
(b) Describe the immediate pilot actions required to recover from vortex ring state.
(c) What are the key indicators that a helicopter is entering auto rotation?

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of auto rotation apply to emergency landing procedures in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min